

effects of an amenity depend on the tradeoffs at the margin—i.e., the costs and benefits faced by firms and workers who are just indifferent to providing or accepting the amenity. Applied to our context, this implies that the wage effects of chatbot encouragement hinge on the marginal firm and worker.

Labor market competition. Who ultimately benefits from the productivity gains of AI chatbots depends crucially on the nature of labor market competition. In competitive labor markets, workers are compensated only for productivity gains not driven by firm-level investments (Acemoglu and Pischke, 1998; Becker, 1964). In this case, worker-level adoption should affect worker wages (Figure 2), whereas firm-level adoption should not. Instead, firm-level adoption should manifest in total employment (Figure 3), as firms move down their product demand curves.

An important alternative is the monopsony model of Card et al. (2018), in which firms exert monopsony power because workers have heterogeneous preferences over employers, but firms cannot price discriminate across workers. In this framework, adopters do not experience within-firm wage differentials; instead, any earnings effects from chatbots arise at the firm level through labor demand adjustments. Thus, while within-firm earnings gaps between adopters and non-adopters (Figure 2) should not emerge in this setup, firm-level outcomes such as total wage bills, employment, and average wages should still be affected (Figure 3). Whether firm-level employment effects translate into wages depends on the elasticity of workers’ labor supply across employers—that is, how heterogeneous their firm preferences are. Finally, even in a monopsony setting, adopters should reallocate toward encouraged firms, as shown in Figure 3.

A.2.2 Formal Setup

This section introduces a formal model of chatbot adoption and labor market outcomes, built on a set of simplifying assumptions for analytical tractability. First, we treat AI chatbots as tools that affect output only when actively deployed by workers. This reflects the current state of chatbot technology—dependent on direct human prompting—while

abstracting from more autonomous systems. Second, we model workers' adoption decisions, as modeled in Section A.1, using a Roy framework that allows for heterogeneity in both productivity benefits and adoption costs. We apply a parallel Roy adoption model to firms' initiatives, assuming that firms encourage chatbot use when expected profit gains exceed encouragement costs. Finally, we assume that product markets are characterized by monopolistic competition and labor markets are perfectly competitive. Modeling employer encouragement as a binary decision, our model falls within the Rosen (1986) framework of amenity provision and equalizing differences.

Productivity and earnings. Worker i 's earnings from working at firm j are the product of a firm-specific skill price W_j and the worker's efficiency units of human capital H_{ij} :

$$Y_{ij}(A_{ij}) = W_j H_{ij}(A_{ij}). \quad (9)$$

Workers are endowed with a unit of baseline human capital, which may be augmented through chatbot use, $A_{ij} \in \{0, 1\}$:

$$H_{ij}(A_{ij}) = \exp(B_{ij}A_{ij}), \quad (10)$$

where B_{ij} captures the productivity benefit of AI chatbots.

As in Section A.1, the productivity benefit and adoption cost depends on worker i 's type and firm j 's chatbot initiatives:

$$B_{ij} = \alpha_{b0} + \alpha_{b1}E_j + \beta_b \times i \quad (11)$$

$$C_{ij} = \alpha_{c0} + \alpha_{c1}E_j + \beta_c \times i. \quad (12)$$

Firm production and demand. Firms produce using efficiency units of labor:

$$Y_j = H_j = \int_{i \in \mathcal{I}(j)} H_{ij} di, \quad (13)$$

where $\mathcal{I}(j)$ denotes the set of workers employed at firm j .

The production function in Equation (13) models AI chatbots as tools that affect output only when actively deployed by workers. While we view this as a realistic representation of current chatbot technology—which depends on direct human prompting—Section A.2.1 examines the implications of a more general technology where chatbots may affect production beyond individual use.

Product markets are characterized by monopolistic competition, where atomistic and horizontally differentiated firms hire labor and produce output to maximize profits. Firms face the product demand curves:

$$Y_j = P_j^{-\epsilon}, \quad (14)$$

where ϵ is the elasticity of demand across firms within the sector.

Labor demand. Labor markets are perfectly competitive, so workers are paid according to the marginal revenue product of labor:

$$W_j = \frac{\partial P_j Y_j}{\partial H_j} = (1 - 1/\epsilon) Y_j^{-\frac{1}{\epsilon}}. \quad (15)$$

Since firms' production and demand curves are symmetric, and workers derive the same net benefit $\hat{\alpha}_1$ from employer encouragement across employers, two skill prices W_j exist in equilibrium: one at encouraged employers W_1 and one at non-encouraged employers W_0 . Let $j \in \{0, 1\}$ indicate whether a firm encourages chatbot use.

Equilibrium skill prices adjust to equate the supply and demand for human capital across encouraged and non-encouraged employers. These supply and demand forces reflect workers' sorting and adoption decisions, as well as firms' encouragement and hiring policies. We characterize equilibrium skill prices in Section A.2.3.

Employer encouragement. Employers derive log utility from flow profits and encourage AI chatbots if the utility from the resulting profit gain exceeds their encouragement

cost κ_j :

$$E_j = \mathbf{1} [\kappa_j \leq \log \Pi_j(1) - \log \Pi_j(0)], \quad (16)$$

where $\Pi_j(E)$ denotes firm j 's flow profits under encouragement policy $E \in \{0, 1\}$. Let F_κ denote the CDF of encouragement costs.

Constant returns to scale (Equation (13)) and CES demand (Equation (14)) imply that a firm's optimal flow profit is a constant fraction $\frac{1}{\epsilon}$ of revenues:

$$\Pi_j = \frac{1}{\epsilon} P_j Y_j = \frac{1}{\epsilon} Y_j^{1-1/\epsilon}. \quad (17)$$

Combining Equations (15) and (17), the profit gain from encouraging chatbots equals:

$$\log \Pi_j(1) - \log \Pi_j(0) = (1 - \epsilon)(\log W_1 - \log W_0) \quad (18)$$

Since the profit gain is common across employers, the encouragement policy follows a cut-off rule based on encouragement costs:

$$E_j = \mathbf{1} [\kappa_j \leq \kappa], \quad \text{with} \quad \kappa = (1 - \epsilon)(\log W_1 - \log W_0). \quad (19)$$

Adoption. Workers derive log-utility from labor earnings and adopt AI chatbots if the monetary benefit exceeds the adoption cost C_{ij} . Fixing the worker's employer, the adoption rule is:

$$A_{ij} = \mathbf{1} [\log Y_{ij}(1) - \log Y_{ij}(0) \geq C_{ij}] \quad (20)$$

$$= \mathbf{1} [B_{ij} \geq C_{ij}] \quad (21)$$

$$= \mathbf{1} [\hat{\alpha}_0 + \hat{\alpha}_1 E_j + \hat{\beta} \times i \geq 0], \quad (22)$$

where $\hat{x} = x_b - x_c$ denotes net benefits, resulting in the adoption rule from Section A.1.2.

This implies a strict cutoff rule for adoption:

$$A_{ij} = \mathbf{1} [i \leq i^*(E_j)], \quad \text{with} \quad i^*(E) = -\frac{1}{\hat{\beta}}(\hat{\alpha}_0 + \hat{\alpha}_1 E). \quad (23)$$